

Going Further with Google CA

Going further with conversational agents

With your baseline competency of conversational agents, you are well placed to explore further and below are some areas for your consideration:

- a. Generators
- b. Improving RAG accuracy
- c. Publish agent

GENERATORS

Generators

1. Generators are reusable LLM prompts which can be called directly from flows or playbooks
2. They are ideal for tasks such as summarizing, transforming text, answering questions, etc.
3. This [video](#) provides a basic illustration on how it can be used
4. For further details, please refer to the documentation [here](#)

IMPROVING RAG ACCURACY

Improving RAG accuracy (Data store settings)

1. Click on Settings
2. Select Generative AI
3. Select Data store to configure the below
 - a. Grounding parameters
 - b. Data store prompt and model selection
 - c. Summarization Model
4. Click on "Create a custom prompt" to fine-tune a prompt which suits your project requirements (if necessary)
5. Reference to the following [documentation](#)
 - a. You would be required to have the following variables in your prompt:
 - \$original-query or \$rewritten-query
 - \$sources
 - \$conversation (Recommended for contextual awareness)
 - instruct the LLM to return "NOT_ENOUGH_INFORMATION"

The screenshot shows the 'Conversational Agents' settings page for 'Project SMU-org' and 'Agent SR Bot'. The interface includes tabs for 'General', 'Generative AI', 'Deterministic Flows', 'Connectivity', 'Speech and IVR', 'UI Settings', and 'Security'. A red box labeled '1' highlights the settings gear icon in the top right. A red box labeled '2' highlights the 'Generative AI' tab. A red box labeled '3' highlights the 'Data store' sub-tab under 'Generative AI'. A red box labeled '3a' highlights the 'Grounding' section, which includes options to 'Enable grounding' (checked), 'Lowest score allowed' (set to 'Low: We have low confidence that the response is grounded'), and 'Apply grounding heuristics filter' (checked). A red box labeled '4' highlights the 'Create a custom prompt' radio button under the 'Summarization prompt' section. Below this, the 'Data store fallback' section shows 'Fallback link' and 'Enable Generative AI' both checked.

Improving RAG accuracy (FAQ Data Store)

1. You can overwrite an answer by adding an FAQ entry for a specific question.
 - a. Using a [FAQ Data Store](#) (must be in CSV)
2. Use the [parse and chunk configurations](#) to optimize responses.
 - a. Experiment with different parser and chunking size

(Note that layout parser is likely the best, so do not need to spend too much time experimenting with this. Same for chunking, exhaustive experimentation is not needed.)

Document parsing

Select a parser to apply to your documents. [Learn more about parsing and chunking documents.](#)

Default document parser *
Layout Parser

Select the parser override for specific document formats:

[+ Add exception for a file format](#)

Layout parser settings

☐ Enable table annotation

☐ Enable image annotation

Document chunking

☒ Enable advanced chunking configuration

Chunk size limit
500 Tokens

☒ Include ancestor headings in chunks

Document parsing

Select a parser to apply to your documents. [Learn more about parsing and chunking documents.](#)

Default document parser *

- Digital Parser
- OCR Parser
- Layout Parser**

Improving RAG accuracy (Multiple Data Stores)

If you have multiple data stores, you can boost selected data stores for higher priority in the RAG response

1. Under AI Applications
 - a. Connected Data Stores
 - b. Click on view
2. Click on Configurations
 - a. Create Control

The first screenshot shows the Google Cloud console interface. In the left sidebar, 'AI Applications' is selected. Under 'Apps', 'Connected data stores' is highlighted with a red box and a red number '1'. The main panel shows a table with one entry: 'SR Bot' (App type: Chat), with a 'view' link highlighted in a red box.

The second screenshot shows the 'Configurations' page for the 'SR Bot' app. In the left sidebar, 'Configurations' is highlighted with a red box. In the main panel, under the 'Controls' section, the '+ Create control' button is highlighted with a red box and a red number '2'.

Improving RAG accuracy (Multiple Data Stores)

1. Enter Control Name
2. Control Type Select “Boost/bury”
3. Select the data store
 - a. Adjust the filters (leave blank to apply to all)
 - b. Adjust boost / bury value of data store.
4. Leave remainder as default.
5. Submit

Define rules and preferences to influence how products appear in search results, giving you precise command over your commerce search.

1 What do you want to do?

Define your control by giving it a clear, memorable name and selecting the control type that determines how it will influence search results.

1

Control name *
Adjust Data Store Weights

Control ID: adjust-data-store-weights. It cannot be changed later. [Edit](#)

2

Control Type *
Boost/bury

[Continue](#)

2 Set Rule Scope & Impact

3a

documents this rule will affect, and by how much it will boost or bury them.

Select a data store *
sustainability-report_1755072734326

Choose where your AI application gets its information (e.g., your list of products, support tickets, customer reviews).

Filter

Tell the rule which specific items to look for. You can combine search terms using AND, OR, NOT, and group terms with (). For exact phrases, use quotes \"\".

category: "electronics" AND price < 500, status: "open" AND priority: "P1"

Specify criteria for the items this rule should apply to. If you leave this blank, the rule will apply to all items in your chosen data store. [Learn more](#).

3b

Boost/bury value

Adjust this slider to make matching items appear more (boost) or less (bury) prominently in your AI application's results. A higher number means a stronger effect.

Value: 0.00

Bury(-1) Boost(+1)

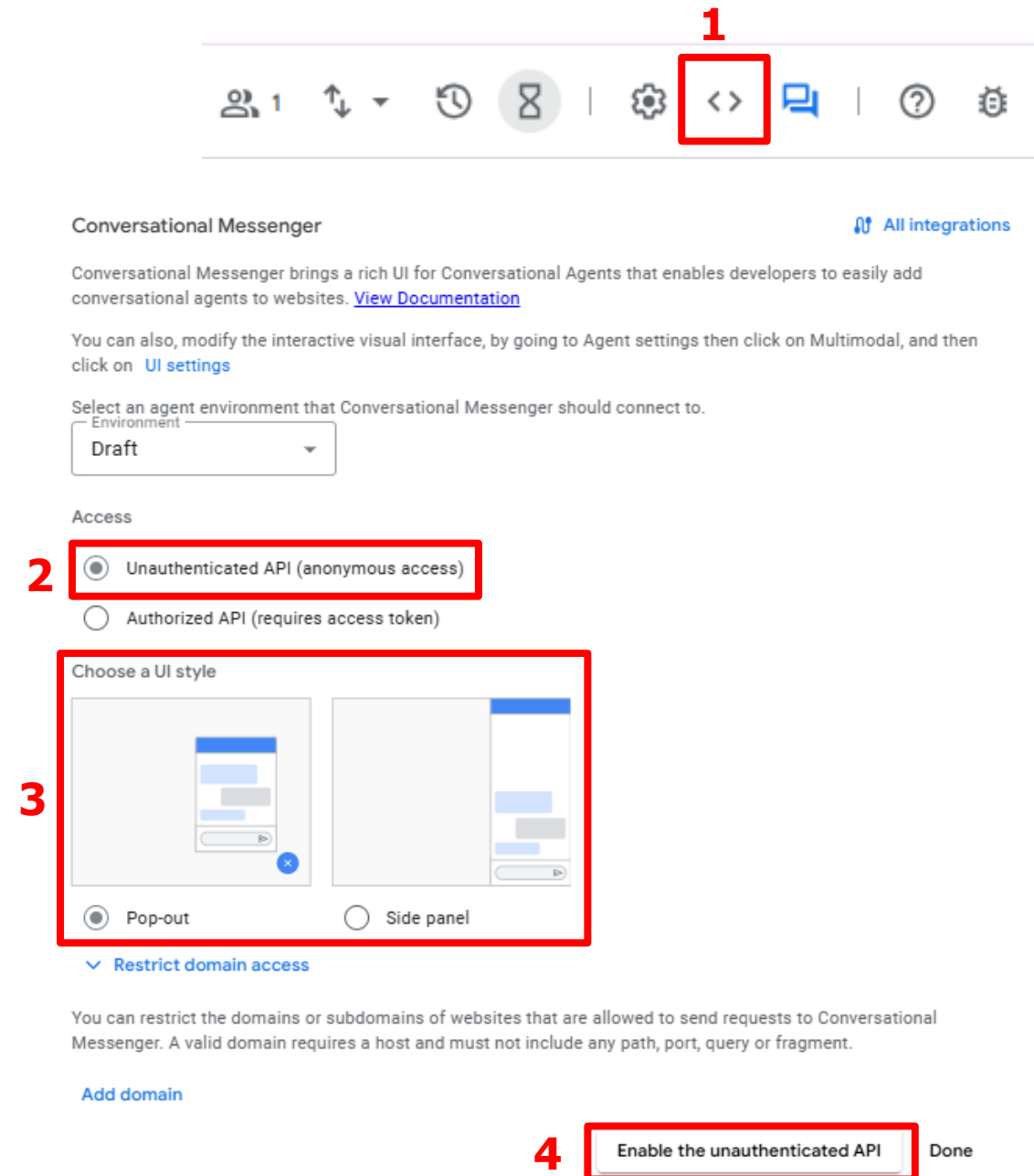
[Continue](#)

PUBLISH AGENT

Publish agent

We are now going to publish your agent so that it can be assessed on the internet.

1. Click on the Publish agent icon at the upper right corner of the console
2. Ensure that Unauthenticated API is selected. This allow anonymous access.
3. Choose your UI style
4. Click on Enable the unauthenticated API



The screenshot shows the 'Conversational Messenger' settings page. At the top right, a toolbar contains several icons; the '<>' icon is highlighted with a red box and the number '1'. Below the toolbar, the 'Environment' dropdown is set to 'Draft'. Under the 'Access' section, the 'Unauthenticated API (anonymous access)' radio button is selected and highlighted with a red box and the number '2'. The 'Choose a UI style' section shows two options: 'Pop-out' (selected) and 'Side panel', both highlighted with a red box and the number '3'. Below this, there is a link to 'Restrict domain access'. At the bottom right, the 'Enable the unauthenticated API' checkbox is checked and highlighted with a red box and the number '4', next to a 'Done' button.

Conversational Messenger [All integrations](#)

Conversational Messenger brings a rich UI for Conversational Agents that enables developers to easily add conversational agents to websites. [View Documentation](#)

You can also, modify the interactive visual interface, by going to Agent settings then click on Multimodal, and then click on [UI settings](#)

Select an agent environment that Conversational Messenger should connect to.

Environment
Draft

Access

☒ Unauthenticated API (anonymous access)

☐ Authorized API (requires access token)

Choose a UI style

☒ Pop-out ☐ Side panel

[Restrict domain access](#)

You can restrict the domains or subdomains of websites that are allowed to send requests to Conversational Messenger. A valid domain requires a host and must not include any path, port, query or fragment.

[Add domain](#)

4 ☒ Enable the unauthenticated API [Done](#)

Publish agent

1. You will now see a window containing html code
2. Click on the copy icon to copy the code

Conversational Messenger

[All integrations](#)

Conversational Messenger brings a rich UI for Conversational Agents that enables developers to easily add conversational agents to websites. [View Documentation](#)

You can also, modify the interactive visual interface, by going to Agent settings then click on Multimodal, and then click on [UI settings](#)

⚠ End-user interactions with the Conversational Messenger widget will be billed to your GCP account.

Add this agent to your website by copying the code below:

1

```
<link rel="stylesheet" href="https://www.gstatic.com/dialogflow-console/fast/df-
messenger/prod/v1/themes/df-messenger-default.css">
<script src="https://www.gstatic.com/dialogflow-console/fast/df-messenger/prod/v1/df-
messenger.js"></script>
<df-messenger
  location="us-central1"
  project-id="concise-ranger-362608"
  agent-id="f10f0940-63b4-4c6e-931c-ac00b8fbb93f"
  language-code="en"
  max-query-length="-1">
  <df-messenger-chat-bubble
    chat-title="SR Bot 2">
  </df-messenger-chat-bubble>
</df-messenger>
<style>
df-messenger {
  z-index: 999;
  position: fixed;
  --df-messenger-font-color: #000;
  --df-messenger-font-family: Google Sans;
}
```

2


Environment: **Draft**

Disable the unauthenticated API

Try it now

Done

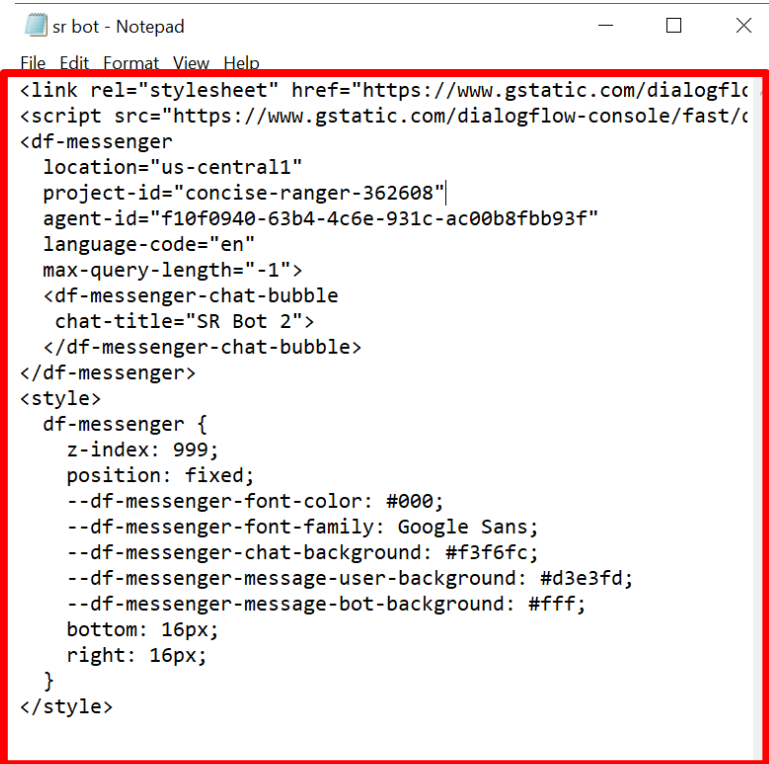
Create html file

In order to publish your agent on a website, you need to create a html file and paste the code you copied from the previous step.

1. Open notepad
2. Paste the html code
3. Save as "sr bot.html"

[It is important to include the double inverted commas when saving the file to ensure that it is saved as a html file]

2



```

File Edit Format View Help
<link rel="stylesheet" href="https://www.gstatic.com/dialogflow-console/fast/c
<script src="https://www.gstatic.com/dialogflow-console/fast/c
<df-messenger
  location="us-central1"
  project-id="concise-ranger-362608"
  agent-id="f10f0940-63b4-4c6e-931c-ac00b8fbb93f"
  language-code="en"
  max-query-length="-1">
  <df-messenger-chat-bubble
    chat-title="SR Bot 2">
  </df-messenger-chat-bubble>
</df-messenger>
<style>
  df-messenger {
    z-index: 999;
    position: fixed;
    --df-messenger-font-color: #000;
    --df-messenger-font-family: Google Sans;
    --df-messenger-chat-background: #f3f6fc;
    --df-messenger-message-user-background: #d3e3fd;
    --df-messenger-message-bot-background: #fff;
    bottom: 16px;
    right: 16px;
  }
</style>

```

3

File name: "sr bot.html"

Upload to bucket

Now, we are going to create a bucket to upload your html file.

1. Enter a globally unique name for your bucket
2. Click **CREATE**
3. Deselect **Enforce public access prevention on this bucket**
4. Click **CONFIRM**

← Create a bucket

- **Get Started**

Pick a globally unique, permanent name. [Naming guidelines](#)

1

Optimize storage for data-intensive workloads ☐

Labels (optional) ☐

[CONTINUE](#)
- **Choose where to store your data**

Location: us (multiple regions in United States)
Location type: Multi-region
- **Choose a storage class for your data**

Default storage class: Standard
- **Choose how to control access to objects**

Public access prevention: On
Access control: Uniform
- **Choose how to protect object data**

Soft delete policy: Default
Object versioning: Disabled
Bucket retention policy: Disabled
Object retention: Disabled
Encryption type: Google-managed

2 [CREATE](#) [CANCEL](#)

Public access will be prevented

This bucket is set to prevent exposure of its data on the public internet.

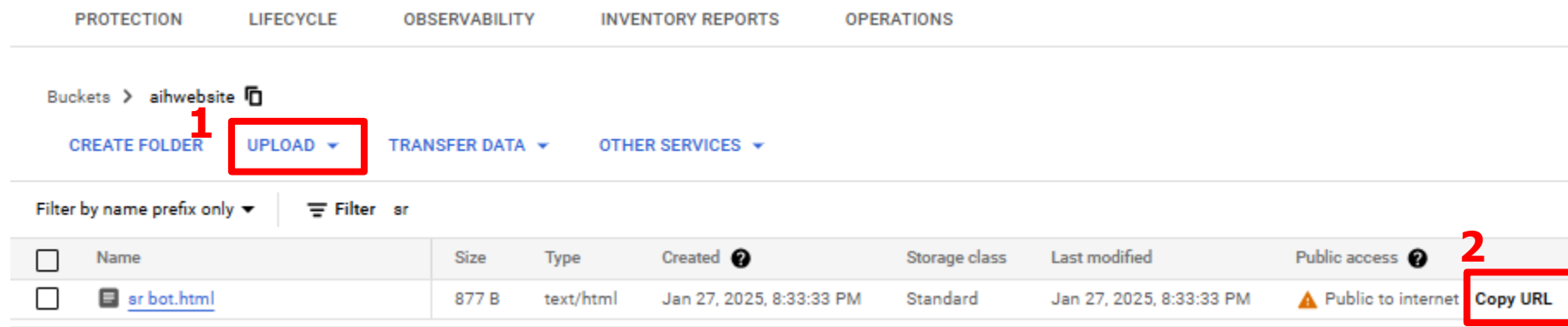
Keep this setting enabled unless you have a use case that requires public access (such as static website hosting). You can change it now or later. [Learn more](#)

3 ☐ Enforce public access prevention on this bucket
☐ Don't show this message again

4 [CANCEL](#) [CONFIRM](#)

Testing

1. After the bucket is created, upload the html file you created which contains the html code you copied from Conversational Agent
2. To test your bot, click on the **Copy URL** icon under the Public access column
3. Paste the URL in a browser and you should see an icon for the agent



OTHERS

Others

1. If you are interested in seeing how RAG works using python code, this [video](#) contains a guided run-through of the chunking, embedding, retrieval and generation process through two source pdfs.